

Underwater sound focusing via quasi-three-dimensional gradient-index lens

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Gradient-index lenses, enabled by meta-atoms, have demonstrated effective focusing capabilities in both airborne and submerged environments. However, the current focusing capabilities remain constrained primarily to two-dimensional systems. Here we introduce a quasi-3D gradient-index lens designed for underwater sound focusing. This lens, characterized by a simple yet effective design, features an additional degree of freedom along the axial direction, enabling controlled wave refraction in longitudinal and lateral directions. To achieve this, eigenfrequency analysis and a scattering parameter retrieval method are used to calculate the band structure and characterize the effective material properties. Numerical simulations and experimental validation confirm the efficacy of the lens in focusing underwater sound and highlight its potential for high-resolution sensing, advanced biomedical imaging, and efficient energy harvesting.

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I. INTRODUCTION

Studying crystal-like and periodic structures at the sub-wavelength scale has provided a robust platform for revealing previously unexplored physical properties [1–4]. In particular, the controlled manipulation of the refractive index has resulted in the realization of real-world systems for metamaterials, metalenses, and gradient-index (GRIN) lenses, including those exhibiting locally resonant characteristics [5–14]. Building on these findings, significant progress has been made in directing acoustic waves in both air and water with the use of GRIN lenses, which are

typically available in two-dimensional (2D) configurations [15–17].

In general, dimensions are defined by the number of axes along which physical quantities are measured independently. While most studies have been centered mainly on 2D structures for sound focusing [18–20], advancements in 3D printing technology and additive manufacturing now enable the creation of complex structures, such as phononic crystal (PC) GRIN lenses and metalenses, thereby extending their functionality to 3D structures. Additionally, the use of high-impedance materials in fabrication extends the applications from airborne to waterborne environments [21–23]. In particular, additive manufacturing has made it possible to produce poroelastic microlattices, broadband flattened Luneburg lenses, and other advanced acoustic components [24–27]. 3D Luneburg lenses, as higher-dimensional systems,

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